INCH-POUND

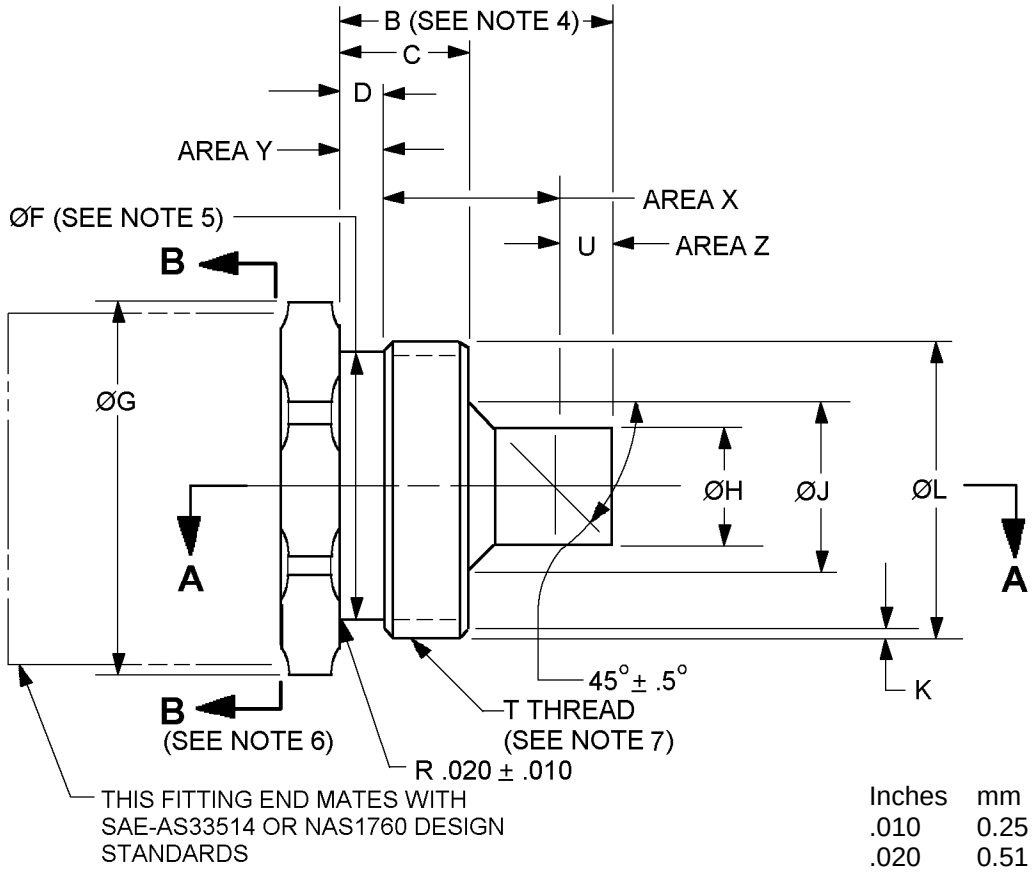
MS27391E
5 February 2009
SUPERSEDING
MS27391D
24 September 2003

DETAIL SPECIFICATION SHEET

NIPPLE, FLARELESS, TUBE TO HOSE - SWIVEL NUT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

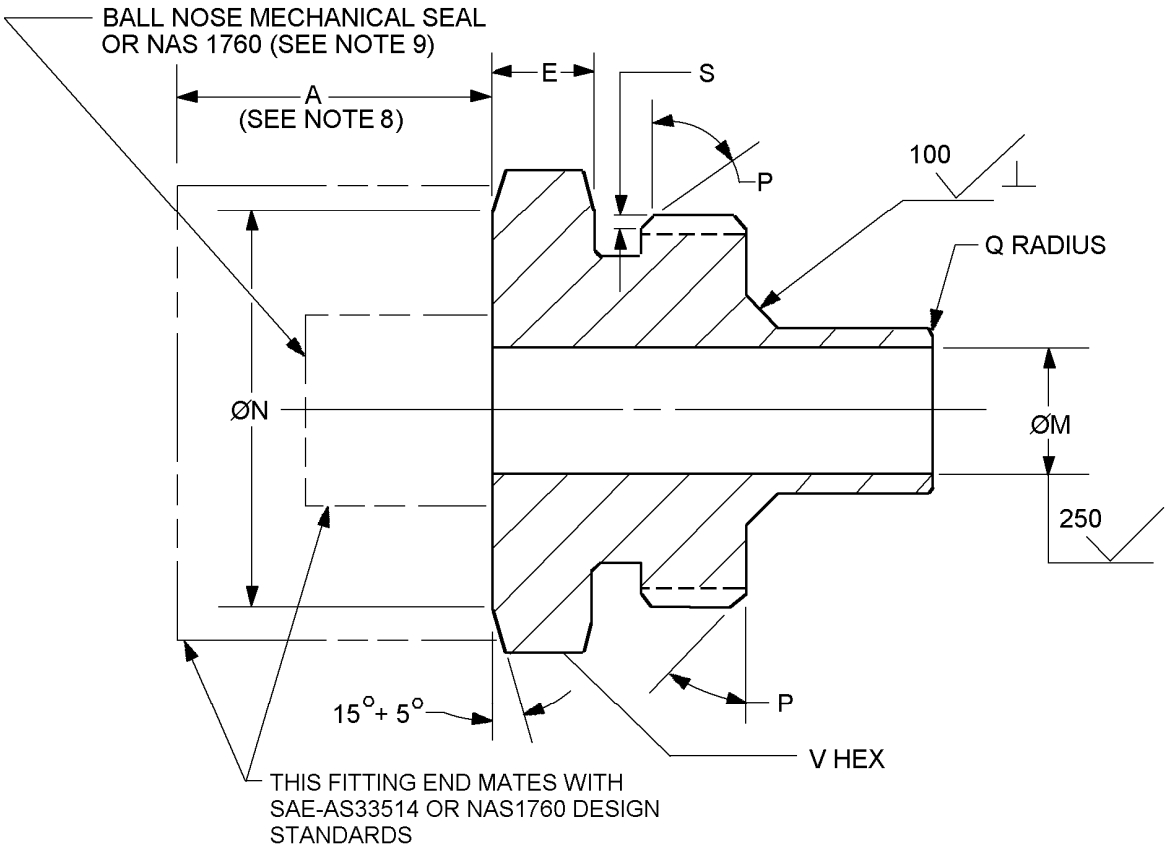
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27272



Inches	mm
.010	0.25
.020	0.51

FIGURE 1. Nipple tube to hose dimensions and configuration.

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SECTION A-A

Dash number and material designator		Hose OD inches (mm)	Tube OD inches (mm)
Steel	Alum		
-3/-4C	---	.188 (4.78)	.250 (6.35)
-4C	---	.250 (6.35)	.250 (6.35)
-5C	---	.313 (7.95)	.313 (7.95)
-6C	---	.375 (9.53)	.375 (9.53)
-8C	-8D	.500 (12.70)	.500 (12.70)
-10C	-10D	.625 (15.88)	.625 (15.88)
-12C	-12D	.750 (19.05)	.750 (19.05)
-16C	-16D	1.000 (25.40)	1.000 (25.40)
-20C	-20D	1.250 (31.75)	1.250 (31.75)
-24C	-24D	1.500 (38.10)	1.500 (38.10)

FIGURE 1. Nipple tube to hose dimensions and configuration - Continued.

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Dash number and material designator		A inches (mm) ±.010 (0.25)	B (see note 5) inches (mm) ±.005 (0.13)		C inches (mm) ±.005 (0.13)		D inches (mm) ±.005 (0.13)	
Steel	Alum		Steel	Alum	Steel	Alum	Steel	Alum
-3/-4C	---	.533 (13.54)	.735 (18.67)	---	.255 (6.48)	---	.075 (1.91)	---
-4C	---	.518 (13.16)		---		---		---
-5C	---	.588 (14.94)	.730 (18.54)	---	.250 (6.35)	---	.085 (2.16)	---
-6C	---	.617 (15.67)		---		---	.050 (1.27)	---
-8C	-8D	.717 (18.21)	.870 (22.10)	.865 (21.97)	.270 (6.86)	.265 (6.73)	.085 (2.16)	.080 (2.03)
-10C	-10D	.824 (20.93)	.950 (24.13)	.957 (24.31)	.300 (7.62)	.307 (7.80)	.100 (2.54)	.107 (2.72)
-12C	-12D	1.028 (26.11)	1.035 (26.29)	1.018 (25.86)	.360 (9.14)	.343 (8.71)		.083 (2.11)
-16C	-16D	1.028 (26.11)	1.120 (28.45)	1.100 (27.94)	.390 (9.91)	.370 (9.40)	.110 (2.79)	.090 (2.29)
-20C	-20D	1.110 (28.19)	1.335 (33.91)	1.341 (34.06)	.400 (10.16)	.406 (10.31)	.085 (2.16)	.091 (2.31)
-24C	-24D	1.367 (34.72)	1.480 (37.59)	1.480 (37.59)	.500 (12.70)	.500 (12.70)	.100 (2.54)	.100 (2.54)

Dash number and material designator		E inches (mm) ±.005 (0.13)		F dia (see note 4) inches (mm)		G dia inches (mm)	
Steel	Alum	Steel	Alum				
-3/-4C	---	.125 (3.18)	---	.400 (10.16)	±.005	.620 (15.75)	±.005
-4C	---		---				
-5C	---		---	.430 (10.92)		.675 (17.15)	+.015 -.000
-6C	---	.135 (3.43)	---	.495 (12.57)		.750 (19.05)	
-8C	-8D	.150 (3.81)	.155 (3.94)	.600 (15.24)		.960 (24.38)	
-10C	-10D	.170 (4.32)	.163 (4.14)	.785 (19.94)	±.010	1.105 (28.07)	
-12C	-12D	.190 (4.83)	.207 (5.26)	.910 (23.11)		1.385 (35.18)	
-16C	-16D	.220 (5.59)	.240 (6.10)	1.277 (32.44)		1.680 (42.67)	
-20C	-20D	.280 (7.11)	.274 (6.96)	1.589 (40.36)		1.979 (50.27)	
-24C	-24D		.280 (7.11)	1.828 (46.43)		2.340 (59.44)	

FIGURE 1. Nipple tube to hose dimensions and configuration - Continued.

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Dash number and material designator		H dia inches (mm) +.000 -.005 (0.13)	J dia inches (mm) ±.005 (0.13)	K inches (mm)	
Steel	Alum				
-3/-4C	---	.210 (5.33)	.295 (7.49)	.031 (.79)	±.005 (0.13)
-4C	---				
-5C	---	.273 (6.93)	.360 (9.14)		
-6C	---	.335 (8.51)	.425 (10.80)	.015 (.38)	max
-8C	-8D	.431 (10.95)	.530 (13.46)	.031 (.79)	±.005 (0.13)
-10C	-10D	.531 (13.49)	.625 (15.88)	.035 (.89)	
-12C	-12D	.655 (16.64)	.760 (19.30)		
-16C	-16D	.905 (22.99)	1.040 (26.42)	.038 (.97)	
-20C	-20D	1.156 (29.36)	1.275 (32.39)		
-24C	-24D	1.406 (35.71)	1.550 (39.37)	.043 (1.09)	

Dash number and material designator		L dia inches (mm)		M dia inches (mm)		N dia inches (mm) .02 (0.51)
Steel	Alum					
-3/-4C	---	.500 (12.70)	+.000/-.006 (+0.00/-0.15)	.161 (4.09)	+.005 (0.13) -.000	.53 (13.46)
-4C	---					
-5C	---	.562 (14.27)	+.000/-.007 (+0.00/-0.18)	.224 (5.69)		.59 (14.99)
-6C	---	.625 (15.88)		.261 (6.63)		.66 (16.76)
-8C	-8D	.750 (19.05)		.345 (8.76)	+.006 (0.15) -.000	.84 (21.34)
-10C	-10D	.875 (22.23)	+.000/-.008 (+0.00/-0.20)	.440 (11.18)		.97 (24.64)
-12C	-12D	1.000 (25.40)		.560 (14.22)		1.22 (30.99)
-16C	-16D	1.375 (34.93)	+.000/-.009 (+0.00/-0.23)	.828 (21.03)		1.47 (37.34)
-20C	-20D	1.688 (42.88)		1.058 (26.87)		1.78 (45.21)
-24C	-24D	1.938 (49.23)	+.000/-.010 (+0.00/-0.25)	1.282 (32.56)	+.005 (0.13) -.000	2.09 (53.09)

Dash number and material designator		P ±5º		Q radius inches (mm)		S inches (mm)	
Steel	Alum	Steel	Alum				
-3/-4C	---	45º	---	.015 (0.38)	+.005 -.000	.031 (0.79)	±.005 (0.13)
-4C	---		---				
-5C	---		---				
-6C	---	30°	---			.015 (0.38)	max
-8C	-8D	45°	30°	.020 (0.51)		.031 (0.79)	±.005 (0.13)
-10C	-10D				±.005	---	---
-12C	-12D			.030 (.76)		---	---
-16C	-16D					---	---
-20C	-20D					---	---
-24C	-24D			.035 (0.89)		---	---

FIGURE 1. Nipple tube to hose dimensions and configuration - Continued.

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Dash number and material designator		T (see note 7)		U inches (mm)		V inches (mm) ±.02
Steel	Alum	Thread	Pitch diameter Inches (mm)	Steel only		
-3/-4C	---	.5000 - 28 UNJEF - 3A	.4768/.4740 (12.110/12.040)	.25 (6.35)	±.05	.56 (14.22)
-4C	---					
-5C	---	.5625 - 24 UNJEF -3A	.5354/.5325 (13.599/13.526)			.62 (15.75)
-6C	---	.6250 - 24 UNJEF -3A	.5979/.5949 (15.187/15.110)			.69 (17.53)
-8C	-8D	.7500 - 24 UNS -3A	.7229/.7198 (18.361/18.283)	.32 (8.13)	±.12	.88 (22.35)
-10C	-10D	.8750 - 20 UNJEF -3A	.8425/.8392 (21.401/21.316)	.35 (8.89)	±.15	1.00 (25.40)
-12C	-12D	1.0000 - 20 UNJEF -3A	.9675/.9641 (24.574/24.488)			1.25 (31.75)
-16C	-16D	1.3750 - 18 UNJEF - 3A	1.3389/1.3353 (34.008/33.917)	.39 (9.91)	±.19	1.50 (38.10)
-20C	-20D	1.6875 - 18 UNJEF - 3A	1.6514/1.6476 (41.946/41.849)	.48 (12.19)	±.28	1.81 (45.97)
-24C	-24D	1.9375 - 16 UN - 3A	1.8969/1.8929 (48.181/48.080)	.50 (12.70)	±.30	2.12 (53.85)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, break or radius all corners .005, +.005, -.000 inches (-0.13, +0.13, -0.00 mm).
4. All diameters within B dimension shall be concentric within .005 inch (0.13 mm) full indicator movement. All other diameters must be concentric within .010 inches (0.25 mm) full indicator movement.
5. Thread gage must enter thread relief (sizes -10 through -24 only).
6. Any design to the left of plane B-B is acceptable provided the dimension A and the requirements of this specification sheet and the procurement specification are met.
7. Threads shall be in accordance with SAE-AS8879 except for -8 and -24 sizes, which shall be in accordance with ASME B1.1. Threads shall be rolled on corrosion-resistant steel only.
8. Dimension A indicates the overall length after mating parts are assembled. The measurement of the overall dimension "A" of the ghosted area is not measured to the end of MS27391, but is measured to the end of the mating part. It is not to the end of the ball nose mechanical seal of MS27391.
9. If NAS 1760 is used, MIL-DTL-25579/1 defines the method to adjust the hose assembly lengths to meet the original hose assembly lengths in MS8000 through MS8009.
10. Remove all burrs and slivers.
11. Unless otherwise specified, surface roughness shall not exceed 125 μin. R_a in accordance with ASME B46.1.

FIGURE 1. Nipple tube to hose dimensions and configuration - Continued.

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REQUIREMENTS

Nipple hose to tube shall be as specified on figure 1.

This part is a component of MS27386. The item is only procured as an integral part of adapter assemblies.

Material:

Corrosion-resistant steel: Class 304, condition A, in accordance with ASTM A276.

Aluminum alloy: 6061-T651, in accordance with SAE-AMS-QQ-A-225/8.

Finish:

Corrosion-resistant steel: Passivate in accordance with SAE-AMS2700, method 1.

Aluminum alloy: Anodize in accordance with MIL-A-8625, type II, dye yellow.

Lubrication:

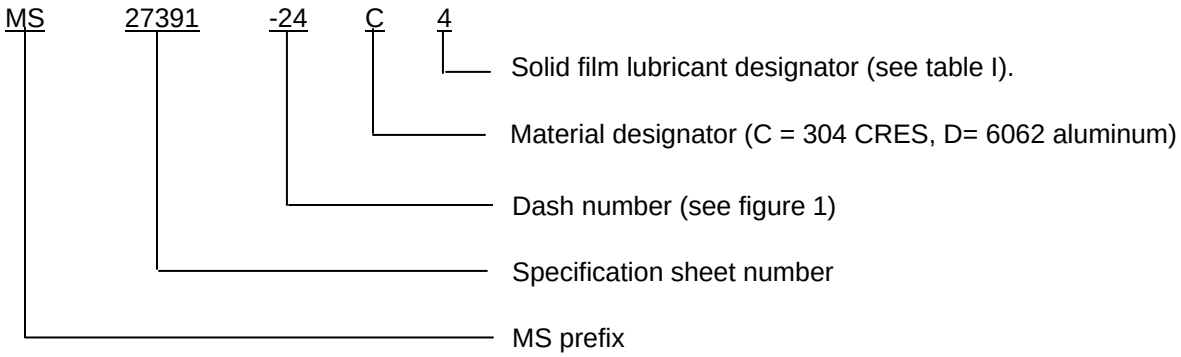
Nipples made with corrosion resistant steel shall be solid-film lubricated in area X with lubricant in accordance with SAE-AS1701, see table I. Solid-film lubrication allowed in area Y, no solid-film lubrication allowed in area Z (see figure 1).

NOTE: Avoid using graphite dry film lubes with aluminum nipples because in a wet environment, graphite becomes corrosive to the aluminum.

TABLE I. Solid film designator.

Dry film designator	SAE-AS1701 class	SAE-AS1701 temperature ranges °F (°C)
Blank	4, 5, or 6	Any temp range
4	4	-65 to +1400°F (-54 to 760°C)
5	5	-65 to +850°F (-54 to 454°C)
6	6	-375 to +850°F (-226 to 454°C)

Part or Identifying Number (PIN):



PIN example: MS27391-24C indicates a nipple hose to tube, size 1.500 inches, corrosion resistant steel, with dry film lubricant temperature rating -65 to +1400°F (-54 to 760°C).

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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-27272, this document references the following:

MIL-A-8625	MS8005	NAS 1760
MIL-DTL-25579/1	MS8006	SAE-AMS-QQ-A-225/8
MS8000	MS8007	SAE-AMS2700
MS8001	MS8009	SAE-AS1701
MS8002	MS27386	SAE-AS8879
MS8003	ASME B1.1	SAE-AS33514
MS8004	ASME B46.1	
MS8008	ASTM A276	

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2008-108)

Review activities:
Army - AT, MI
Navy - MC, SA, SH
Air Force -71

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